Triple Integral over a Cylinder, Two Vertical Planes, and a Horizontal Plane

Stewart, *Calculus: Early Transcendentals*, Section 15.7, Exercise 18

Problem

Evaluate the triple integral.

$$\iiint_E z \, \mathrm{d}V,$$

where E is bounded by the cylinder $y^2 + z^2 = 9$ and the planes $x = 0$, $y = 3x$, and $z = 0$ in the first octant.

Part (a)

This exercise has a single part. The whole solution is Part (a), and the supporting files are named with `part1`.

The symbols. E is the solid region of $\mathbb{R}^3$ that the four surfaces enclose. $\mathrm{d}V$ is the volume element, that is, the volume of a very small piece of E. The symbol z is used in two roles in this exercise. It is the integrand, so it measures the height of a point above the xy-plane, and it is also the third coordinate of a point (x, y, z) of E. Because $\mathrm{d}V$ is positive and $z \geq 0$ on E, the integral is the volume of E weighted by z.

The letter D denotes the projection of E onto the yz-plane. Inside that plane we use polar coordinates: r is the distance from the origin of the yz-plane, so $r^2 = y^2 + z^2$, and θ is the angle measured from the positive y-axis toward the positive z-axis. Then $y = r\cos\theta$ and $z = r\sin\theta$, and the area element of D is $\mathrm{d}A = r \, \mathrm{d}r \, \mathrm{d}\theta$. This exercise uses no spherical coordinates (ρ, θ, φ); the symbol θ above is always the polar angle in the yz-plane.

Step 1. Describe the solid

The cylinder $y^2 + z^2 = 9$ has its axis on the x-axis and its radius is 3. The condition "in the first octant" means $x \geq 0$, $y \geq 0$, and $z \geq 0$. The plane $z = 0$ is the floor of the solid, and the plane $x = 0$ is one vertical wall. The plane $y = 3x$ is vertical because z does not appear in its equation; we write it as $x = y/3$.

A point (x, y, z) is in E exactly when the three conditions below hold:

$$y^2 + z^2 \le 9, \qquad y \ge 0,\ z \ge 0, \tag{1}$$

$$0 \le x \le \frac{y}{3}. \tag{2}$$

Condition (1) is the quarter disk of radius 3 in the yz-plane, because the cylinder keeps the inside and the first octant keeps $y \ge 0$ and $z \ge 0$. Condition (2) is a pair of vertical planes, and it gives the range of x. For a fixed point (y, z) of the quarter disk, the value of x is free from 0 to $y/3$.

Therefore the projection of E onto the yz-plane is the quarter disk

$$D = \{(y, z) : y^2 + z^2 \le 9,\ y \ge 0,\ z \ge 0\}, \tag{3}$$

and the solid is

$$E = \{(x, y, z) : (y, z) \in D,\ 0 \le x \le y/3\}. \tag{4}$$

Figure 1 shows the four faces of E. Figure 2 shows the projection D and the second projection of E onto the xy-plane.

Because $y \le 3$ on D, the largest value of x on E is $3/3 = 1$. So E lies inside the box $[0, 1] \times [0, 3] \times [0, 3]$. This box is used in the numerical check in Step 6.

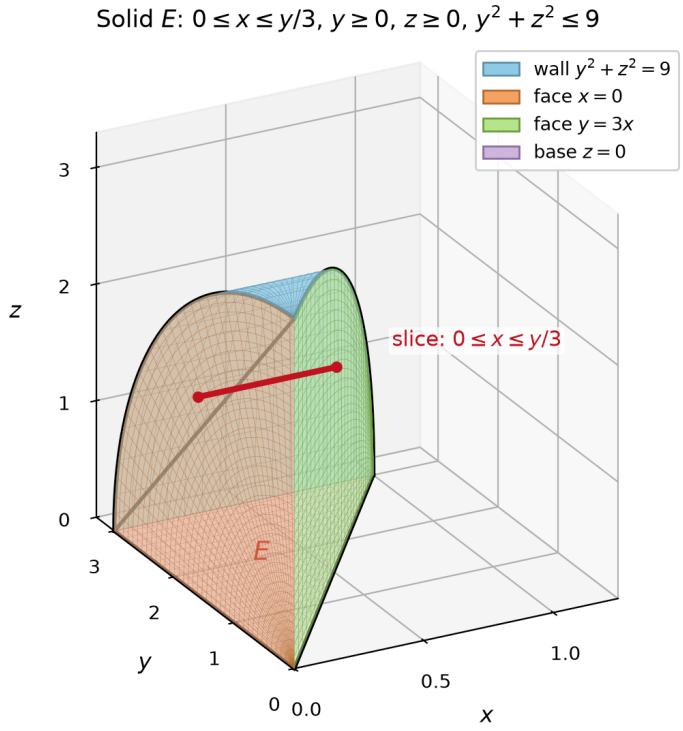


Figure 1: The solid E of Exercise 15.7.18. The curved face is the part of the cylinder $y^2 + z^2 = 9$ that lies between the planes $x = 0$ and $y = 3x$. The flat faces are the disk $x = 0$, the slanted face $y = 3x$, and the base $z = 0$. The red segment is the slice of E at one fixed pair (y, z), and it runs from $x = 0$ to $x = y/3$.

Step 2. Choose the order of integration

Look at Equation (2). The variable x appears only in that pair of inequalities, and both bounds for x are functions of y alone. So a straight line parallel to the x-axis enters E at $x = 0$, leaves E at $x = y/3$, and meets E in exactly one segment for every interior point (y, z) of D. This is the definition of a solid that is simple with respect to x.

Therefore we may integrate with respect to x first, and the two outer integrals run over the quarter disk D:

$$\iiint_E z \, \mathrm{d}V = \iint_D \left(\int_0^{y/3} z \, \mathrm{d}x \right) \mathrm{d}A. \tag{5}$$

This order is legal by Fubini's theorem, because the integrand z is continuous on E and the bounds are continuous functions of (y, z) on D.

Step 3. Compute the inner integral

The variable of the inner integral is x, so z is a constant during that integration. Hence

$$\int_0^{y/3} z \, \mathrm{d}x = z \Big[x \Big]_{x=0}^{x=y/3} = \frac{yz}{3}. \tag{6}$$

The antiderivative of the constant z with respect to x is zx. At the upper limit the value is $z \cdot (y/3)$, and at the lower limit it is 0. The result is nonnegative, and it is 0 only where $y = 0$ or $z = 0$, which are the two edges of E on the coordinate planes.

Step 4. Convert the outer integral to polar coordinates

Insert Equation (6) into Equation (5):

$$\iiint_E z \, \mathrm{d}V = \iint_D \frac{yz}{3} \, \mathrm{d}A. \tag{7}$$

The region D of Equation (3) is a quarter disk, and the integrand $yz/3$ is a product of the two coordinates. Both facts call for polar coordinates in the yz-plane. With

$$y = r\cos\theta, \qquad z = r\sin\theta, \qquad \mathrm{d}A = r \, \mathrm{d}r \, \mathrm{d}\theta, \tag{8}$$

the quarter disk becomes the rectangle

$$0 \le r \le 3, \qquad 0 \le \theta \le \frac{\pi}{2}, \tag{9}$$

because $r^2 = y^2 + z^2 \le 9$ gives $r \le 3$, and the conditions $y \ge 0$, $z \ge 0$ of the first octant give the first quadrant $0 \le \theta \le \pi/2$. Figure 2 shows this quarter disk together with one small polar element of area $r \, \mathrm{d}r \, \mathrm{d}\theta$.

The integrand becomes

$$\frac{yz}{3} = \frac{(r\cos\theta)(r\sin\theta)}{3} = \frac{r^2 \sin\theta\cos\theta}{3}. \tag{10}$$

The product of Equations (8) and (10) is the full element of the integral:

$$\frac{yz}{3}\,dA = \frac{r^2\sin\theta\cos\theta}{3}\cdot r\,dr\,d\theta = \frac{r^3\sin\theta\cos\theta}{3}\,dr\,d\theta. \tag{11}$$

One extra power of r appears here. Forgetting that power is the most common error in this type of problem.

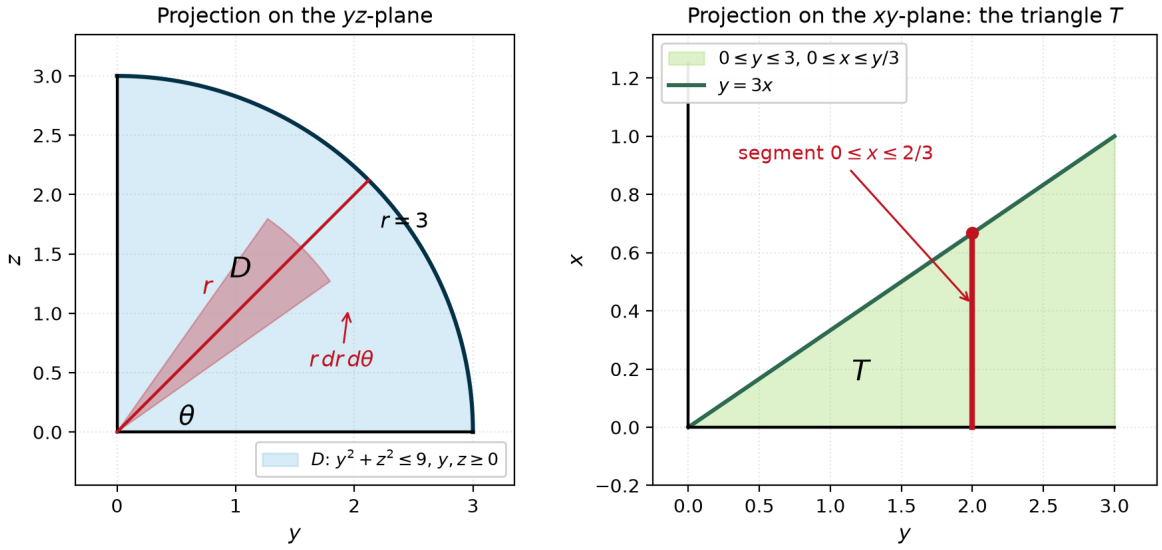


Figure 2: The two projections used in the solution. Left: the projection D of E onto the yz-plane is the quarter disk $0 \le r \le 3$, $0 \le \theta \le \pi/2$. The shaded sector is one element of area $dA = r\,dr\,d\theta$. Right: the projection of E onto the xy-plane is the triangle $0 \le y \le 3$, $0 \le x \le y/3$. The red segment is the slice at $y = 2$, and it runs from $x = 0$ to $x = 2/3$.

Step 5. Evaluate the polar integral

Equation (11) separates into a product of a function of r and a function of θ. Therefore the double integral is a product of two single integrals:

$$\iiint_E z\,dV = \left(\int_0^3 \frac{r^3}{3}\,dr\right)\left(\int_0^{\pi/2} \sin\theta\cos\theta\,d\theta\right) \tag{12}$$

$$= \left(\frac{1}{3}\cdot\frac{3^4}{4}\right)\left(\frac{1}{2}\right) \tag{13}$$

The first integral uses the antiderivative $r^4/4$, so its value is $(1/3)(81/4) = 27/4$. The second integral uses the substitution $u = \sin\theta$, $du = \cos\theta\,d\theta$; the antiderivative is $\sin^2\theta/2$, and its value on $[0, \pi/2]$ is $1/2 - 0 = 1/2$.

The product is

$$\iiint_E z\,dV = \frac{27}{4}\cdot\frac{1}{2} = \frac{27}{8}. \tag{14}$$

Answer

$$\boxed{\iiint_E z \, \mathrm{d}V = \frac{27}{8} = 3.375}$$

Step 6. Check the value

The value was checked twice. The first check is exact and uses SymPy. It evaluates the integral in a different order, so it uses different antiderivatives. The second check is numerical and uses NumPy only.

Exact check in the order z, then x, then y. Take z as the inner variable. On E the bound for z comes from the cylinder: for a fixed y the disk keeps $0 \le z \le \sqrt{9-y^2}$. Then

$$\int_0^{\sqrt{9-y^2}} z \, \mathrm{d}z = \frac{9-y^2}{2}. \tag{15}$$

Next integrate in x over $0 \le x \le y/3$; the integrand (15) does not contain x, so that step multiplies the result by $y/3$:

$$\int_0^{y/3} \frac{9-y^2}{2} \, \mathrm{d}x = \frac{y}{3} \cdot \frac{9-y^2}{2} = \frac{9y-y^3}{6}. \tag{16}$$

Finally integrate in y over $0 \le y \le 3$:

$$\int_0^3 \frac{9y-y^3}{6} \, \mathrm{d}y = \frac{1}{6}\left[\frac{9y^2}{2} - \frac{y^4}{4}\right]_0^3 = \frac{1}{6}\left(\frac{81}{2} - \frac{81}{4}\right) = \frac{1}{6} \cdot \frac{81}{4} = \frac{27}{8}. \tag{17}$$

Equation (17) agrees with Equation (14). The two calculations use different orders of integration, so they are independent checks of the limits and of the antiderivatives.

Numerical check. The three estimates in Table 1 use the exact value $27/8 = 3.375$ as the reference. The two-dimensional Simpson rule integrates Equation (7) in polar coordinates with 300×300 panels. The Monte Carlo method samples the box $[0,1] \times [0,3] \times [0,3]$ and counts the samples that lie in E. The midpoint rule sums z over 300^3 small cells.

Table 1: Numerical confirmation of the exact value $27/8$. The error column compares each estimate with 3.375.

Method	Estimate	Absolute error
Two-dimensional Simpson rule (300×300)	3.375000000225	2.3×10^{-10}
Monte Carlo, $3 \cdot 10^6$ samples	3.367270	7.7×10^{-3}
Midpoint voxel sum, 300^3 cells	3.387274	1.2×10^{-2}
Exact value $27/8$	3.375000000000	—

The three numerical estimates agree with $27/8$ to the accuracy that each method can reach. A geometric estimate supports the magnitude of the answer. The same steps with the integrand 1 in

place of z give the volume of E:

$$V(E) = \int_0^3 \frac{r^2}{3}\,\mathrm{d}r \int_0^{\pi/2} \cos\theta\,\mathrm{d}\theta = \frac{1}{3}\cdot 9 \cdot 1 = 3, \tag{18}$$

because the inner integral in x has length $y/3 = r\cos\theta/3$. The average value of z on E is then

$$\bar{z} = \frac{1}{V(E)} \iiint_E z\,\mathrm{d}V = \frac{27/8}{3} = \frac{9}{8} = 1.125, \tag{19}$$

and this average is plausible, because z runs from 0 to 3 on E and the solid is wider near the floor than near the top.

The scripts `part1_step1.py` to `part1_step4.py`, `part1_figure1.py`, and `part1_figure2.py` produce every number and both pictures in this document, and each script ends with the line `PASS`.

How to solve problems of this kind

1. **Read each surface and classify it.** A surface with no x in its equation, such as $y^2 + z^2 = 9$, is a cylinder whose axis is the x-axis. A surface with no z in its equation, such as $y = 3x$, is a vertical plane. A surface with no constant term, such as $z = 0$, is a coordinate plane. The class of each surface tells you which coordinate is free.

2. **Find the projection first.** Collect the inequalities that do not involve the inner variable. Here the cylinder and the two conditions of the first octant give the quarter disk D in the yz-plane. The projection is always the region where the two outer integrals live.

3. **Test simplicity along the inner axis.** Draw a line parallel to the candidate inner axis. If the line meets E in one segment for every interior point of the projection, then that axis gives the inner integral. Here every line parallel to the x-axis meets E in one segment, because $0 \le x \le y/3$.

4. **Use the shape of the projection to pick the coordinates.** A disk or a quarter disk in the yz-plane calls for polar coordinates in that plane. A rectangle calls for Cartesian coordinates. A solid whose description uses $\rho^2 = x^2 + y^2 + z^2$ calls for spherical coordinates.

5. **Write the volume element of the chosen system.** In polar coordinates of the yz-plane the element is $r\,\mathrm{d}r\,\mathrm{d}\theta$. The factor r is part of the geometry, not part of the integrand, and it must be written before any antiderivative is taken.

6. **Integrate from the inside out and keep the variables straight.** When the inner variable does not appear in the integrand, as in Equation (6), the inner integral is just the integrand times the length of the interval. Say this in words, and then write the multiplication.

7. **Check with a second order and a number.** A second exact order exercises different antiderivatives. A numerical method such as Simpson, Monte Carlo, or a midpoint sum exercises the limits. Compare both, and compare the magnitude with a rough geometric estimate.